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Satellite Monitoring for Air Quality and Health

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Abstract

Data from satellite instruments provide estimates of gas and particle levels relevant to human health, even pollutants invisible to the human eye. However, the successful interpretation of satellite data requires an understanding of how satellites relate to other data sources, as well as factors affecting their application to health challenges. Drawing from the expertise and experience of the 2016–2020 NASA HAQAST (Health and Air Quality Applied Sciences Team), we present a review of satellite data for air quality and health applications. We include a discussion of satellite data for epidemiological studies and health impact assessments, as well as the use of satellite data to evaluate air quality trends, support air quality regulation, characterize smoke from wildfires, and quantify emission sources. The primary advantage of satellite data compared to in situ measurements, e.g., from air quality monitoring stations, is their spatial coverage. Satellite data can reveal where pollution levels are highest around the world, how levels have changed over daily to decadal periods, and where pollutants are transported from urban to global scales. To date, air quality and health applications have primarily utilized satellite observations and satellite-derived products relevant to near-surface particulate matter <2.5 µm in diameter (PM_{2.5}) and nitrogen dioxide (NO₂). Health and air quality communities have grown increasingly engaged in the use of satellite data, and this trend is expected to continue. From health researchers to air quality managers, and from global applications to community impacts, satellite data are transforming the way air pollution exposure is evaluated.

1. INTRODUCTION

Over the past decade, satellite data have increasingly been applied to public health and air quality analyses. These applications have been supported by investments into Earth-observing satellites, as well as initiatives to spur effective utilization of satellite data by the health and air quality communities. While health-relevant, near-surface concentrations of pollutants may be inferred from some satellite data products, the successful interpretation of satellite data for health exposure requires an understanding of the strengths and limitations of satellite data. In particular, health applications require linking satellite data to surface-level air quality concerns.

This review presents an overview of satellite data applications to health and air quality issues, with a variety of recent examples. These insights draw from the work of the NASA HAQAST (Health and Air Quality Applied Sciences Team), a four-year applied research initiative working to connect NASA resources and expertise with stakeholders in the air quality and public health communities (1). The authors of this review are all members and project leaders of the 2016–2020 HAQAST, and present here an overview of key ideas in linking satellite data with air quality and health. (For information on the current HAQAST team, funded from 2021 to 2025, the reader is invited to visit <https://HAQAST.org>).

The strengths and limitations of satellite data are best understood in the context of air quality data more broadly. Three main data sources may be used to characterize ambient concentrations of air quality: in situ measurements, computer models, and remote sensing platforms like satellites. The temporal and spatial scale of analysis fundamentally affects the utility of air quality data, as does the pollutant of interest. The appropriateness of air quality data products for time and space scales is shown in **Figure 1** (2).

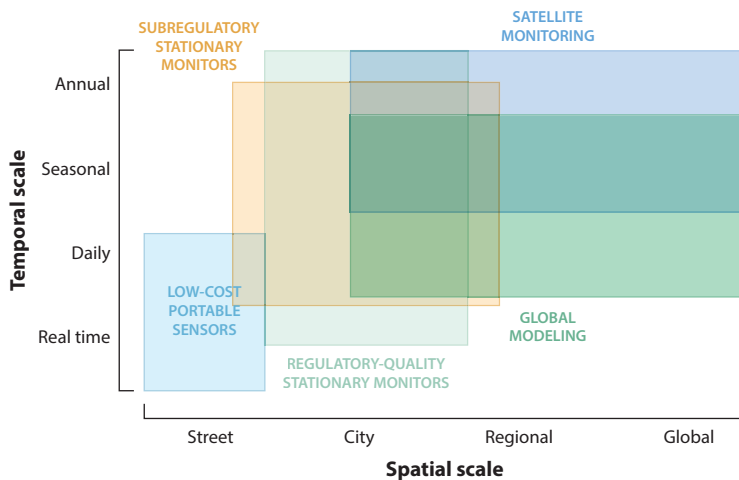


Figure 1

Illustrative summary of general abilities of air pollution monitoring technologies by effective spatial and temporal scales to provide air quality information. Figure adapted with permission from Reference 2; copyright 2019 American Thoracic Society.

Stationary monitors represent the most widely used source of air quality information in most developed countries. These monitors are often considered the gold standard for air quality information and have been widely used for epidemiologic studies. However, ground-based monitors are limited in their geographic coverage, due both to their initial expense and to the cost of maintaining a network. As such, they are typically sited to capture air pollution in urban areas, with less coverage in rural areas. For example, the United States has thousands of monitors for ground-level ozone and particulate matter $<2.5\ \mu\text{m}$ in diameter ($\text{PM}_{2.5}$), although approximately 80% of U.S. counties lack even a single monitor (3). This lack of spatial coverage limits the evaluation of rural exposure to air pollution and impedes the evaluation of nonurban emission sources, from agriculture to forest fires. These spatial biases limit air pollution exposure data available to assess health impacts over a wide area.

Around the world, many countries have little or no publicly available monitoring data, even for common pollutants. Recently, 141 countries were found to have no regular $\text{PM}_{2.5}$ monitoring at all, and only 23 nations have more than three monitors per million inhabitants (4). Shaddick et al. have estimated that 92% of the world's population lives in areas exceeding World Health Organization guidelines for $\text{PM}_{2.5}$ (5). People living in low- and middle-income countries are disproportionately burdened with the mortality associated with outdoor air pollution, and the annual number of deaths is projected to more than double by 2060 (6). de Sherbinin et al. have highlighted the potential for properly constructed satellite-based indicators to inform global air pollution levels for decision-making and public outreach (7).

The primary advantage of satellite data is their high spatial coverage, bridging spatial gaps in current ground-based networks. Satellite data show us where pollution levels are highest around the world, how those levels are changing over the last few decades, and where pollution is transported downwind, including over regional and intercontinental distances. Although many satellite data products may be used to assess year-to-year change, challenges occur in connecting satellite trends to those at the surface and—for some instruments and retrievals—in accounting for changes in instrument functionality over the years.

The most significant constraint of satellite data for health applications is the lack of surface-level information. Satellite data typically report chemical abundance in the column of air above the earth, and additional analysis steps are required to link these columns with air quality on the ground. Most satellites that detect air pollution are polar orbiting, also known as satellites in low-Earth orbit. These satellites pass over the poles of the earth and provide data over most of the world every day, or every few days. While these satellites provide global coverage, they offer just one or two snapshots a day (or less, depending on the instrument design, orbit, cloud cover, and land cover). Recent and upcoming geostationary satellites observing air quality offer the potential for greater temporal data coverage. Geostationary satellites orbit with the earth, so they are able to provide data over a particular geographic region at hourly or better frequencies during daylight hours (8). Depending on the instrument and chemical retrieval, the spatial resolution of satellite data varies from ~ 1 to ~ 50 km. Only a few health-relevant pollutants in the atmosphere may be reliably detected from space, and even these vary in data quality and spatial resolution.

Satellite-based estimates of $PM_{2.5}$ have emerged as some of the most important tools for health assessment, while also some of the most complex applications of satellite data. Concentrations of $PM_{2.5}$ cannot be directly observed from satellites, but rather inferred using models or monitoring data. Rather than detecting $PM_{2.5}$ directly, satellites measure aerosol optical depth (AOD) and other markers of particulate loading in the atmosphere. AOD is a unitless measure of light absorption by particles in the atmosphere, dependent on the total mass of particles in the atmosphere, the size and chemical characteristics of the particles, the vertical distribution of particles the atmosphere, and other factors. As a result, the relationship between columnal AOD and surface $PM_{2.5}$ varies in space and time.

Combining ground monitors with satellite or modeling data can significantly enhance the spatial coverage of the derived $PM_{2.5}$ data, providing continuous datasets in space and time (e.g., 5, 9, 10). Over the years, various methods have been used for deriving this relationship, such as linear regression models (11–13), multiple regression models, or generalized additive models (14–17), as well as more advanced data fusion (18) and machine learning (ML) approaches (19). Compared with parametric regression models, ML algorithms have more relaxed requirements on the independence of observations and collinearity among predictor variables. ML algorithms can also more efficiently handle massive datasets, nonlinear relationships, and interactions among predictors than statistical models. A few widely used ML algorithms include decision tree-based models such as random forest (20–25), gradient boosting (26), neural networks (12), and deep learning (27, 28), as well as ensemble models integrating multiple ML algorithms (29, 30). The temporal resolution of these models ranges from daily to monthly, the study period ranges from several months to multiple years, and the spatial scale ranges from subregional to national. When used to estimate ambient $PM_{2.5}$ concentrations, two recent comparison studies indicated that ML algorithms can often outperform chemical transport models (CTMs) and traditional statistical models such as kriging and multivariate regression, with cross-validation R^2 values exceeding 0.8 when compared with ground observations (31, 32). However, one study with a specific focus on models' spatiotemporal predictive capabilities (comparing models developed using the same set of input data sources) found that statistical models accounting for spatial dependence, such as the downscaler, outperform any other method (33). The underperformance of ML algorithms such as random forest and neural networks might be due to the smaller number of predictors included in this comparison than in typical applications of these methods and the fact that these ML methods do not account explicitly for the spatial dependence of $PM_{2.5}$.

Various ancillary data may be included for improving the relationship function, such as temperature, relative humidity, wind field, land use, and road density. Satellite-derived $PM_{2.5}$ can be further evaluated against ground monitor data, optimizing the AOD– $PM_{2.5}$

relationship for various locations. One example is SPARTAN (Surface Particulate Matter Network; <https://www.spartan-network.org>), which provides public datasets for satellite data evaluation (34). Aircraft data can also support the linking of satellite columns and surface $\text{PM}_{2.5}$ abundance, including from flight campaigns designed to support air quality analysis (e.g., 35).

Past review articles have outlined the opportunities and challenges in applying satellite data for air quality (8), for evaluating emission inventories (36), and for estimating near-surface $\text{PM}_{2.5}$ (10). Here we expand on this body of work to focus on health applications of satellite-derived air quality data, including a discussion of satellite data for epidemiological studies (Section 2) and health impact assessments (Section 3); air quality trends (Section 4); air quality regulation (Section 5); and smoke from wildfires, prescribed burns (PBs) (Section 6), and other emissions sources (Section 7). We conclude with an overview of satellite capabilities and limitations (Section 8).

2. SATELLITE DATA FOR EPIDEMIOLOGICAL STUDIES

Epidemiological studies for air pollution analyze associations between exposure to airborne contaminants and health outcomes, controlling for confounding factors, to isolate and quantify relationships between the concentration of a pollutant and associated health risks. A growing body of epidemiological research at regional to continental scales has used satellite-derived exposure data to quantify health risks of $\text{PM}_{2.5}$ and NO_2 . Past satellite missions were not designed with health analysis in mind, but research connecting remote sensing and epidemiology has highlighted the opportunities to leverage existing data for new health applications. These successes motivate the development of MAIA (Multi-Angle Imager for Aerosols), an upcoming instrument from NASA explicitly designed to quantify the health impacts of different types of air pollution.

Satellite-derived $\text{PM}_{2.5}$ exposure estimates enable air pollution evaluation in areas where monitors are sparse or nonexistent. Using space-based measurements can improve estimation of spatiotemporal changes in $\text{PM}_{2.5}$ to support improved understanding of health response to pollution exposure. For example, Strickland et al. reported significant associations between satellite-derived daily $\text{PM}_{2.5}$ exposure and emergency room visits for several pediatric conditions, including asthma or wheeze and upper respiratory infections, in the U.S. state of Georgia (37). Combining satellite remote sensing data and CTM simulations, Stowell et al. estimated the associations between cardiorespiratory acute events and exposure to smoke $\text{PM}_{2.5}$ in Colorado and found statistically significant associations for asthma [odds ratio (OR) = 1.081] and combined respiratory disease (OR = 1.021) for an increase in fire smoke $\text{PM}_{2.5}$ of $1 \mu\text{g}/\text{m}^3$ (38; see also 39). Xiao et al. examined the associations between maternal $\text{PM}_{2.5}$ exposure and adverse birth outcomes using three exposure assessment methods and found that health effects calculated using gap-filled satellite $\text{PM}_{2.5}$ had similar magnitudes to those using central-site measurements, but with narrower confidence intervals due to reduced exposure error (40). At the national scales, satellite-derived short-term increase in $\text{PM}_{2.5}$ exposure has been associated with an increase in daily mortality in the Medicare population in the United States, and long-term average $\text{PM}_{2.5}$ exposure has been positively linked to elevated risk of cardiovascular disease and incident stroke among Chinese adults (41–43).

Satellite-derived $\text{PM}_{2.5}$ is becoming an important tool to advance air pollution health effects research in developing countries with sparse ground monitoring networks. For example, Tapia et al. reported that satellite-derived $\text{PM}_{2.5}$ exposure was associated with adverse pregnancy outcomes, cardiorespiratory emergency room visits, and daily cardiorespiratory mortality, especially among older people, in Lima, Peru (44–46). Heft-Neal et al. reported that an increase of $10 \mu\text{g}/\text{m}^3$ in $\text{PM}_{2.5}$ concentration is associated with a 9% rise in infant mortality in sub-Saharan Africa,

suggesting a doubling of previous estimates of global deaths of infants that are associated with air pollution (47).

Satellite observations have also been instrumental in estimates of health effects of NO₂ (48). As with other pollutants, NO₂ from satellites can enable air pollution evaluation in areas where monitors are sparse or nonexistent. While the satellite detects the column of NO₂ abundance rather than surface concentrations, studies have found a correlation in spatial and temporal patterns between column and surface NO₂ (49, 50). NO₂ concentrations are highly spatially variable, with concentrations that decline to urban background levels at a distance greater than 300–500 m from major roadways (51). Further, asthma development may be more strongly associated with intraurban variation in NO₂ concentrations than with urban-rural variation (52, 53). Therefore, for NO₂, very high spatial resolution concentration estimates (100 m to 1 km) are desirable for epidemiological analysis. Exposure assessment methods using satellite data combined with techniques such as land use regression modeling are increasingly able to predict NO₂ concentrations at these fine spatial scales (54). With higher spatial resolutions available from recent and upcoming satellites, assessing NO₂ exposure for epidemiological analysis is rapidly advancing in availability and accuracy.

Common sources that contribute to PM_{2.5}, both emitted directly and formed chemically, include the combustion of fossil fuels, biomass burning (both PBs and wildfires), and biogenic emissions such as from forests, agricultural soils, and the ocean. These sources produce a complex mix of reactive sulfur, nitrogen, and carbon that create ozone and PM_{2.5}, with strong spatial and temporal variations in their relative importance. The upcoming MAIA mission from NASA is designed to support future epidemiological studies on the relative toxicity of these different particulate matter (PM) components. As the first NASA mission designed explicitly for public health analysis, MAIA builds upon the legacy of MISR (Multi-Angle Imaging SpectroRadiometer). MAIA will have the capability to inform the chemical speciation of PM_{2.5} to support species-specific health assessments. The MISR instrument defines a set of aerosol mixtures to represent aerosol types around the world representing the major ambient particle species. MISR-retrieved aerosol properties have been successfully used as predictors of major PM_{2.5} species such as sulfate, nitrate, organic carbon (OC), and elemental carbon (EC) (55–58). The upcoming MAIA instrument is similarly designed to combine multispectral, polarimetric, and multiangular capabilities into an imaging system capable of mapping total and speciated PM at neighborhood scale over a selected set of urban centers around the world (59, 60). Over the primary target areas, MAIA's sampling frequency will be three to four times per week. In addition to various aerosol optical properties, the MAIA research team will combine satellite data and models to generate ground-level daily PM_{2.5} sulfate, nitrate, OC, EC, and dust concentrations at 1-km spatial resolution. MAIA is being coordinated with planned epidemiological studies to help determine the relative toxicity of various PM components and to assess the effects of particle size and composition on adverse birth outcomes, cardiovascular and respiratory disease, and premature death.

3. SATELLITE DATA AND HEALTH IMPACT ASSESSMENTS

Health impact assessment (or risk assessment) is a process for estimating the burden of disease attributable to air pollution in a given population. These studies combine concentration–response functions from epidemiological studies with estimates of population exposure to air pollutants. Health risk assessments may also be used to estimate the changes in the disease burden under different exposure conditions (e.g., future climate change or emissions policy implementation). A key challenge for air pollution health impact assessments, from local to global scales, is the need for measurements or model predictions of air pollutant concentrations that are high quality,

standardized, and continuous over long periods, with sufficiently high spatial resolution to capture variability commensurate with population gradients. Increasingly, satellite remote sensing data are being used in combination with in situ monitors and CTM, through data fusion or other methods, to estimate air pollution-related health impacts.

Satellite-derived estimates of surface-level $\text{PM}_{2.5}$ concentrations have been incorporated into a variety of health impact assessments that use the high spatial resolution, broad spatial coverage, and year-to-year availability that are intrinsic to these datasets. Remote sensing-based estimates of exposure from van Donkelaar et al. (61) were used by Evans et al. (62) to estimate the global premature deaths associated with long-term exposure to $\text{PM}_{2.5}$. This type of satellite-informed data product has become a standard component of global scale estimates of $\text{PM}_{2.5}$ health impacts that are updated annually (63). Satellite-derived data, often resolved at the 1- to 10-km scale, have been used to downscale coarser global model calculations of $\text{PM}_{2.5}$ in order to better capture sharp gradients in population distributions. This approach increases the ability of global models to inform health impacts from individual $\text{PM}_{2.5}$ sources, sectors, or future emission scenarios (see **Figure 2**) (64–67).

Recent assessments have highlighted the potential for satellite data to inform public health risks from ambient air pollution. Satellite-derived $\text{PM}_{2.5}$ from Dalhousie University (9, 10, 68, 69) were used for global estimates of $\text{PM}_{2.5}$ in the 2017 Global Burden of Disease study (GBD) and other studies (70). The GBD health burden estimates are derived using health impact assessment methods, which combine information on exposure-response functions (ERFs) for specific diseases and causes of death, baseline rates of those health outcomes, estimates of ambient air pollution concentrations, and the numbers of people exposed. The $\text{PM}_{2.5}$ ERFs used by the GBD study are based on integrating ERFs from a range of epidemiologic literatures, including those of ambient air pollution, household air pollution, second hand smoking, and active smoking (71). The GBD assumes that particle toxicity is independent of PM source type and depends only on concentration, consistent with the bulk of the epidemiological evidence for premature mortality risk due to long-term $\text{PM}_{2.5}$ exposure.

The widely varying $\text{PM}_{2.5}$ exposure levels around the world pose a challenge in estimating global impacts of $\text{PM}_{2.5}$. Risk estimates used in the GBD drew primarily from cohort studies in the United States and Europe, where $\text{PM}_{2.5}$ concentrations seldom exceed 25 ug/m^3 on an annual average. Recently, however, new cohort studies have emerged from China, and over time, it can be anticipated that relevant studies will be published from India and other countries where concentrations range up to 100 ug/m^3 and beyond, thus providing empirical information on the ERF in a range that, until now, has only been estimated (72, 73). In fact, one study focusing on the ERF at high concentrations estimated 8.9 (7.5–10.3) million deaths globally from $\text{PM}_{2.5}$ in 2015 (71), which is markedly higher than GBD 2017's estimate of 2.9 (2.5–3.4) million (70).

High-quality satellite-derived estimates of $\text{PM}_{2.5}$ have only been produced in the past decade, which has raised awareness of ambient air pollution as an issue that spans the local and the global scales (74, 75). While air quality has improved over the past few decades in, for example, the United States and western Europe, driven primarily by clean air regulations, pollutant concentrations in China have increased markedly and have likely peaked and then declined in recent years (76–78). In contrast, concentrations continue to grow and are expected to continue to worsen in, for example, India and Africa (79).

Fewer health impact assessments have focused on NO_2 , despite the growing utilization of satellite-detected NO_2 for other air quality applications. An example of a global NO_2 health impact assessment has found that traffic-related NO_2 exposure has been associated with approximately four million pediatric asthma cases globally each year, about 13% of the global asthma burden (80). This percentage ranged up to 48% in urban areas, where NO_2 is highest. NO_2

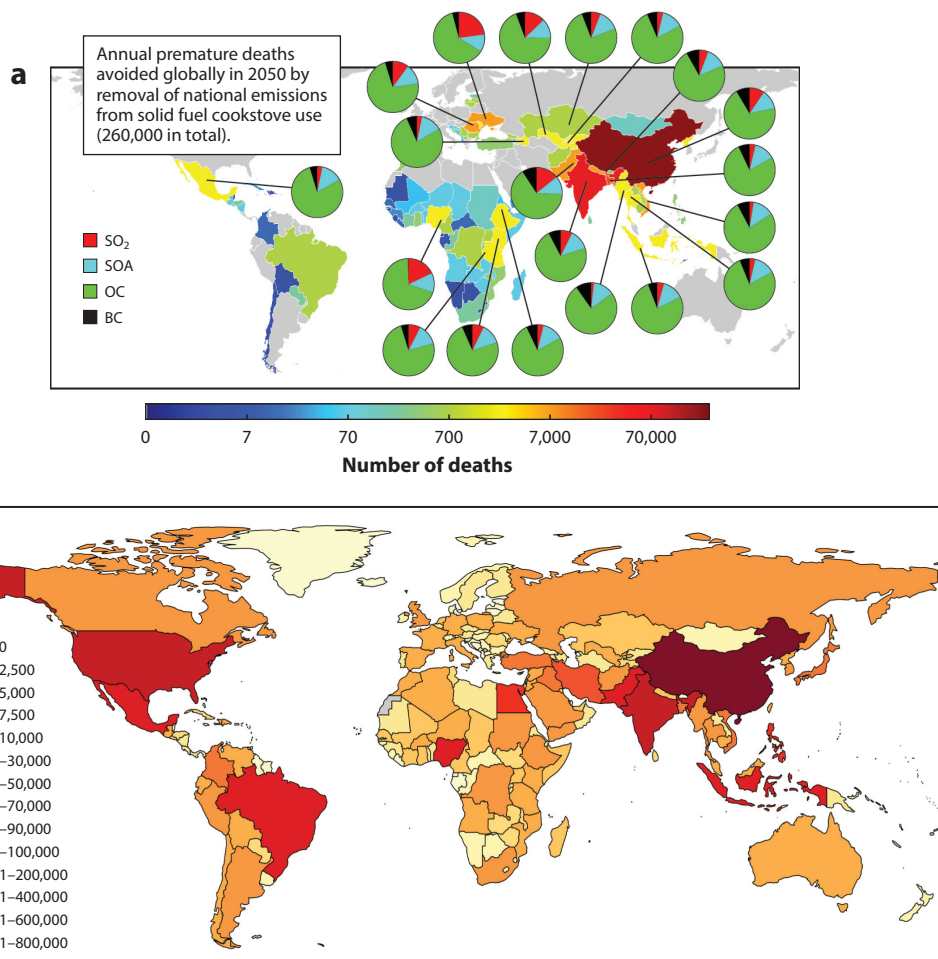


Figure 2

(a) Model simulations in Reference 67 conducted at $2^\circ \times 2.5^\circ$ used to estimate PM_{2.5} health impacts owing to emissions from particular species and countries, with satellite-derived PM_{2.5} estimates used to downscale the simulation results to $0.1^\circ \times 0.1^\circ$ resolution. Panel adapted with permission from Reference 67; copyright 2017 the authors. (b) Number of new pediatric asthma cases in each country due to NO₂ exposure in 2011. Panel adapted with permission from Reference 80; copyright 2019 the authors. Abbreviations: BC, black carbon; NO₂, nitrogen dioxide; OC, organic carbon; PM_{2.5}, particulate matter $<2.5 \mu\text{m}$ in diameter; SO₂, sulfur dioxide; SOA, secondary organic aerosol.

exposure and pediatric asthma incidence is now under consideration for inclusion in the GBD study, where satellite data would be a key input for exposure assessment. If this risk–outcome pair is included, it would complement the current estimates of PM_{2.5}- and ozone-attributable mortality burdens, and it would provide a fuller accounting of the health damages from ambient air pollution worldwide (70).

4. SATELLITE DATA AND AIR QUALITY TRENDS

Satellite data show that pollution levels in urban areas typically change slowly over time (i.e., years), although the construction of power plants or the implementation of emission controls

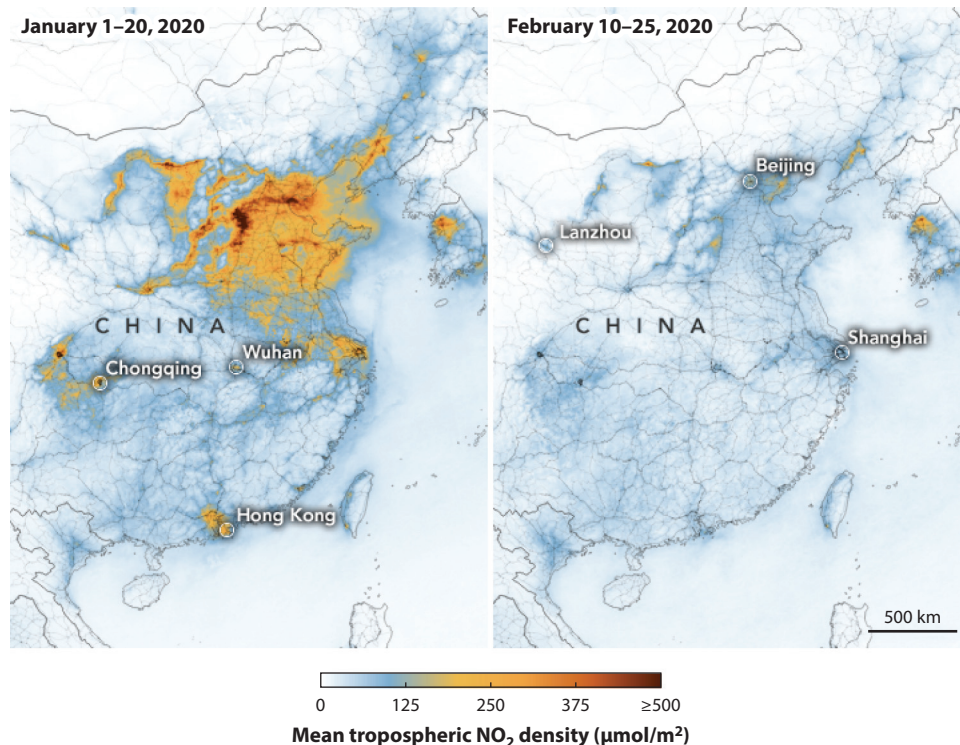


Figure 3

Nitrogen dioxide (NO₂) columns over eastern China in January 2020 (*left*) and February 2020 (*right*) retrieved from the Ozone Monitoring Instrument (OMI) aboard the Aura satellite. Changes reflect reduced economic activity in response to the COVID-19 (coronavirus disease 2019) pandemic. Images courtesy of NASA (<https://airquality.gsfc.nasa.gov/news/airborne-nitrogen-dioxide-plummets-overchina>).

on power plants can rapidly change pollutant levels (81). Records from the Ozone Monitoring Instrument (OMI) aboard the Aura satellite indicate that urban NO₂ levels have decreased by 20–60% over much of the United States and Western Europe since 2005 and over China since about 2011. These trends are due to emission controls on power plants and transportation. In contrast, economic growth in many tropical and subtropical countries has led to increased pollution levels over this same period (82–84).

Satellite data indicate that the recent lockdowns associated with the COVID-19 (coronavirus disease 2019) pandemic have led to decreases in many pollutants over several months in the boreal winter and spring of 2020 (see **Figure 3**). In Chinese cities, NO₂ columns decreased 40% relative to the same time in the previous year, with somewhat lower reductions (20–38%) in the United States and Western Europe (85). However, understanding the changes in air quality resulting from these emissions reductions requires considering the time of year at which the reductions occurred, the background pollution levels in the region, and the role of weather in both photochemical production and the removal of pollutants through precipitation.

While satellites cannot yet measure ozone directly at the surface, they can provide valuable information on ozone in the mid-troposphere (6). Ozone is not emitted directly but is produced from precursor gases such as nitrogen oxides (NO_x = NO + NO₂) and volatile organic compounds (VOCs) that react in the presence of sunlight. Satellite measurements of NO₂ and formaldehyde

(HCHO), an indicator of reactive VOCs, provide an indicator of chemical regime and a valuable resource for evaluating and improving the models used for understanding the sources of ozone. As emissions decline in some parts of the world and continue to increase in others, the mixture of ozone sources at a given location will change with time (86).

Air quality data from satellites, including for trend analysis, bear relevance to a range of policy and regulatory issues. Currently, the U.S. Environmental Protection Agency (EPA) National Ambient Air Quality Standards (NAAQS) only consider ground-based monitor data when determining compliance with the standards, although in a recent evaluation of compliance with the sulfur dioxide (SO₂) standard, models were used in the absence of monitors to evaluate NAAQS compliance. While the U.S. regulatory framework is tied directly to ground-based monitoring networks, satellite data are playing a growing role in providing evidence of spatial and temporal trends in atmospheric pollutant transport, chemistry, and emissions. The air pollution regulatory process recognizes the potential for natural sources to affect compliance with air quality standards and thus includes mechanisms to screen out high-pollution events that are not caused by regulated emissions. Source attribution is required for all high-pollution events, as the specific mechanisms depend on the source type (for example, pollution produced from natural sources versus transport of pollution produced from international anthropogenic emissions).

Because satellite measurements intrinsically sample the column of air above Earth's surface, direct comparisons to air quality stations operated near the surface remain challenging. However, the spatial coverage of satellite data complement ground-based networks to support policy applications. For example, long-term satellite records provide clear evidence for the success of air quality regulations at reducing two trace gases, SO₂ (as shown in **Figure 4**) and NO₂. Each of these is associated with direct health impacts and contributes to chemically formed pollution: SO₂ contributes to sulfate PM and NO₂ contributes to both to ozone and to nitrate PM. In addition to interannual trends, satellites can detect trends for species that have strong seasonal emissions. **Figure 5** shows the highly varying seasonal patterns of NH₃, another precursor of sulfate and nitrate aerosols, due to agricultural emissions from crop fertilization and livestock waste. Despite its importance in PM_{2.5}, NH₃ is not regulated in the Clean Air Act in part due to a lack of observational constraints on emissions and concentrations. Satellite NH₃ measurements are helping to guide the placement of future sites in ground-based networks for quantifying nitrogen deposition (87) and constraining emissions (88). Numerous satellite instruments also detect HCHO, a useful indicator for chemical reactions involving VOCs that lead to ozone and organic aerosol formation (89). During the warm season over the United States, the dominant HCHO source is isoprene, a biogenic VOC emitted from vegetation. New techniques show promise for direct retrieval of isoprene, which, together with HCHO, will provide important constraints on the atmospheric chemistry of ozone and haze formation (90). In summary, a growing portfolio of satellite data products can be applied to improve both the spatial and temporal aspects of emission inventories, as well as to provide key insights regarding atmospheric processes affecting health-relevant pollutants.

Satellite data are now incorporated in some aspects of U.S. air quality policy implementation, and we give some brief examples of these applications here. The annual U.S. EPA air quality trends report now includes long-term changes in tropospheric column NO₂ observed from space (91). Under the NAAQS, states with counties out of compliance with a standard are required to submit a state implementation plan (SIP) that describes an approach to attaining the standard. Satellite NO₂ data have been featured in SIPs submitted by the state of Texas for declining levels of NO_x, a key precursor to ozone formation, and satellite NO₂ and HCHO data have been used by the state of Connecticut as weight-of-evidence for photochemical conditions shaping regional high-ozone events (92–94). Satellite data have been applied to identify episodic, high-pollution events

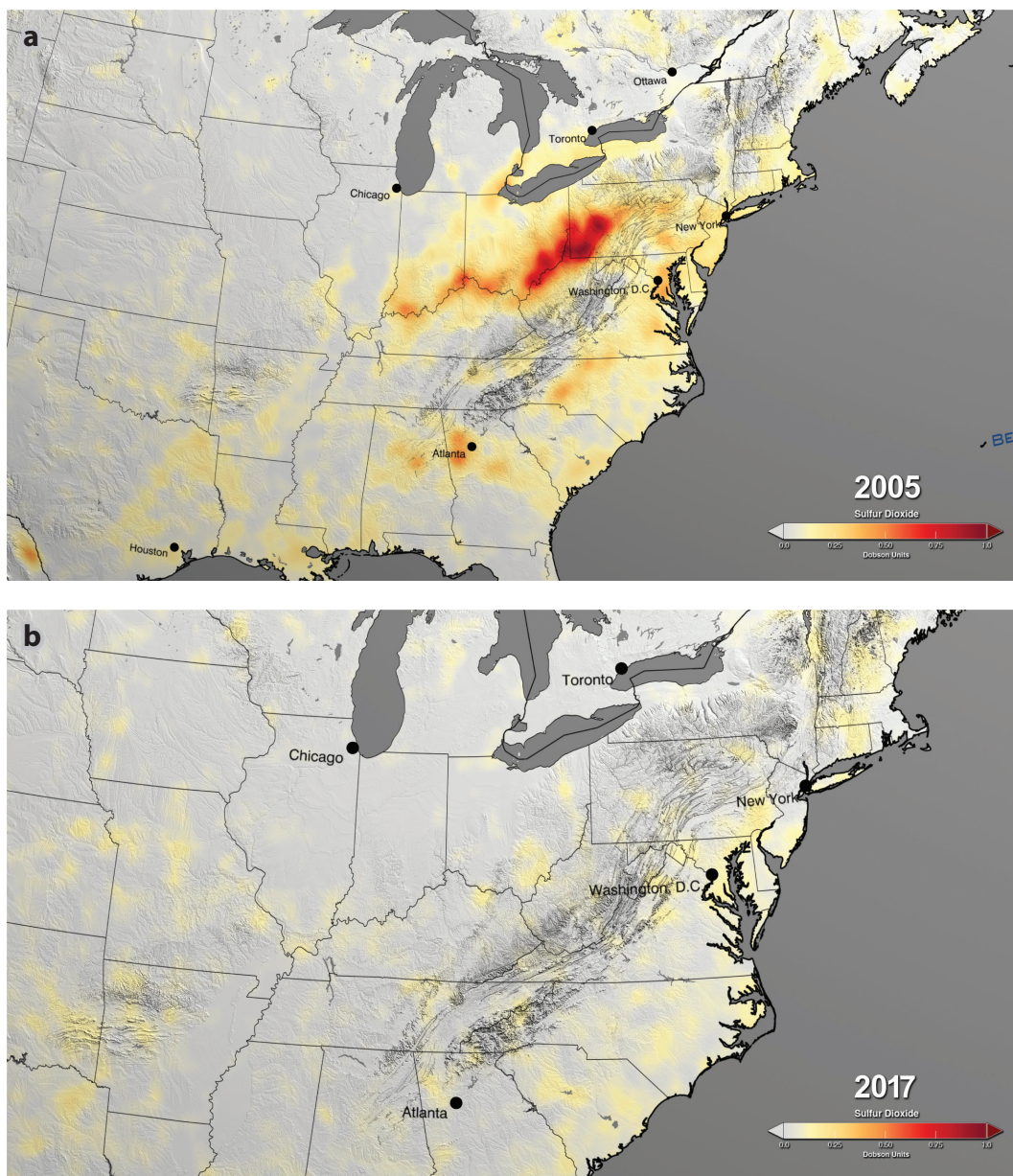


Figure 4

Sulfur dioxide (SO_2) columns over the eastern United States in 2005 (*a*) and in 2017 (*b*) retrieved from the Ozone Monitoring Instrument (OMI) aboard the Aura satellite. Changes reflect technologies, policies, and fuel use changes. Images courtesy of NASA (<https://airquality.gsfc.nasa.gov/particulate-matter>).

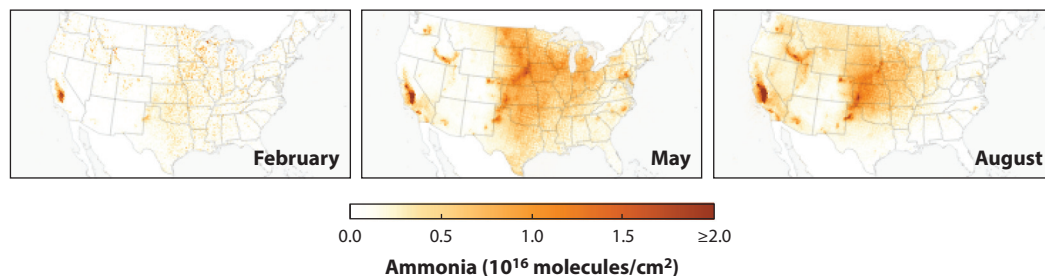


Figure 5

Ammonia (NH₃) columns over the United States in 2016 for February (*left*), May (*middle*), and August (*right*), showing increased agricultural emissions from crop fertilization and animal waste. Data retrieved from the infrared atmospheric sounding interferometer instruments onboard the MetOp satellites. Images courtesy of NASA (<https://earthobservatory.nasa.gov/images/144351/the-seasonal-rhythms-of-ammonia>).

from sources such as wildfires, dust, and even fertilizer application in agricultural regions, often in combination with models and ground-based measurements. More specifically, multiple satellite products have been used in exceptional event demonstrations that make a case for excluding these events from counting toward noncompliance with the ozone NAAQS by documenting a range of observations (weight of evidence) indicating that ozone enhancements were produced from transported wildfire plumes rather than anthropogenic emissions (95). Satellite products have also been used in combination with models and other observations to identify exceptional events associated with high-ozone air masses from the stratosphere into the lower troposphere, as well as international transport of ozone pollution (96).

5. SATELLITE DATA FOR WILDFIRE RESPONSE AND PRESCRIBED BURNS

In the United States, wildfires and PBs (collectively referred to as wildland fires) are becoming an ever-larger fraction of health-damaging air emissions. Remotely sensed data have a wide variety of applications to both PB and wildfire operations, giving information about the atmospheric aerosol loading, trace gases emitted from fires, and especially where the fires are occurring via satellite fire detections (97). While fires are a natural part of many of our ecosystems, smoke from wildland fires is a significant health issue (98–102). PM_{2.5} levels during fires can rise to levels of thousands of micrograms per cubic meter, and ozone levels can rise by tens of parts per billion, even on days outside of the ozone season (103, 104). For both pollutants, the resulting levels routinely exceed the U.S. NAAQS, and states are identifying cases where fires have led to high levels as part of their air quality planning activities (e.g., as part of exceptional event demonstrations) (105).

Satellite data products inform us about where the fires are occurring and horizontal and vertical transport patterns so that we can protect ourselves from the smoke with the aid of smoke forecasting systems and decision support programs such as the Interagency Wildland Fire Air Quality Response Program (106–111). Fundamental to such systems are the satellite fire detection products from the Moderate Resolution Imaging Spectroradiometer (MODIS), Visible Infrared Imaging Radiometer Suite (VIIRS), and Advanced Baseline Imager instruments aboard NASA and NOAA (National Oceanic and Atmospheric Administration) polar-orbiting and geostationary satellites. These instruments detect thermal anomalies across multiple wavelengths to determine a temperature and radiative energy per pixel, which are used to calculate a fire radiative power (FRP). The resolutions of these products range from 375 m to 4 km, with temporal resolutions of twice

daily to every 5 min. By detecting fires on the landscape we can then calculate PM emissions with FRP (e.g., the NASA Fire Energetics and Emissions Research) and estimate how high smoke is lofted into the atmosphere (112) using, for example, the FRP-based method by Sofiev et al. (113). These data also give insight into the seasonality and widespread occurrence of wildland fires. As new satellites are launched, new methods are being developed to aggregate satellite fire detections from multiple platforms using high-temporal resolution geostationary fire detection data to simulate timing of fire activity (a combination of fire front movement and internal perimeter burning) to allocate emissions hourly, simulating, for example, the explosive early-morning fire growth of the 2018 Camp wildfire (114). New methods are also available to statistically combine satellite observations, modeled concentrations, and ground measurements to provide best estimates of smoke concentrations during a wildfire event; this data fusion has been shown to better represent concentrations than any single source of concentration data (115).

Visible satellite imagery and AOD give information about the extent of plumes and the total loading of aerosols in the atmosphere. However, the vertical distribution of smoke is important, as it determines how much smoke is near the surface where people breathe. Instruments such as the CALIOP (Cloud Aerosol Lidar with Orthogonal Polarization) satellite lidar system on the NASA CALIPSO (Cloud Aerosol Lidar and Infrared Pathfinder Satellite Observations) satellite help inform these questions by measuring vertically allocated backscattering along a narrow satellite path to provide a vertical distribution of particulates in the atmosphere (116). Other products such as MISR and the MAIAC (Multi-Angle Implementation of Atmospheric Correction) algorithm developed for MODIS estimate the height of the smoke plume's top (117, 118).

Air quality and health agencies, as well as technical specialists deployed with wildfire-fighting incident management teams, use the aforementioned smoke forecasting systems, surface monitoring data, and satellite products available through tools such as NASA Worldview to issue smoke forecasts and advisories to the public. One such product, the Smoke Outlook, is initialized with a statistical model based on surface observations, MODIS AOD, as well as meteorological parameters (119). The Smoke Outlook and other information are posted to smoke blogs such as the blog California Smoke Information (<http://californiasmokeinfo.blogspot.com/>), which gets millions of hits during periods of wildfire smoke impacts. The EPA also incorporates satellite fire detections and analyzed smoke plumes, along with surface monitoring data from permanent and temporary monitoring systems, to inform the public about current smoke and fire conditions in their Fire and Smoke Map (<https://fire.airnow.gov/>). Since satellite data products can be daunting to new users, we have developed two online 15-min training videos, "The Basics of Satellite Data for Smoke and Fire" and "NASA Worldview for Fire," to bring these important data to a new user community (<https://www.airfire.org/projects/haqast/2017NorthernCAWildfiresTT/training>).

PB of dead and live vegetation for agricultural, land clearing, or silvicultural purposes, or simply to reduce the wildfire risk, is the single largest source of primary PM in the Southeast United States (98). Most PB occurs in the late winter and early spring and is made up of many small (10–100 acres) fires (**Figure 6b**). Individual fires can lead to locally high pollutant levels, and collectively they can lead to a substantial regional background of smoke-based pollution that has been associated with adverse health outcomes, potentially because PM from PBs shows an increased oxidative potential (103, 120–123). A growing air quality management concern is the expected increase in prescribed fire activity, in part to address the increasing risk of wildfires.

The ability to manage the location and time of PBs provides leverage to forest and air quality managers to reduce the adverse impacts of both prescribed and wildfire emissions, as PBs can be planned for places and times that minimize exposure. The Southern Integrated Prescribed Fire Information System (<https://sipc.ce.gatech.edu/SIPFIS/map/>) is a WebGIS (geographic

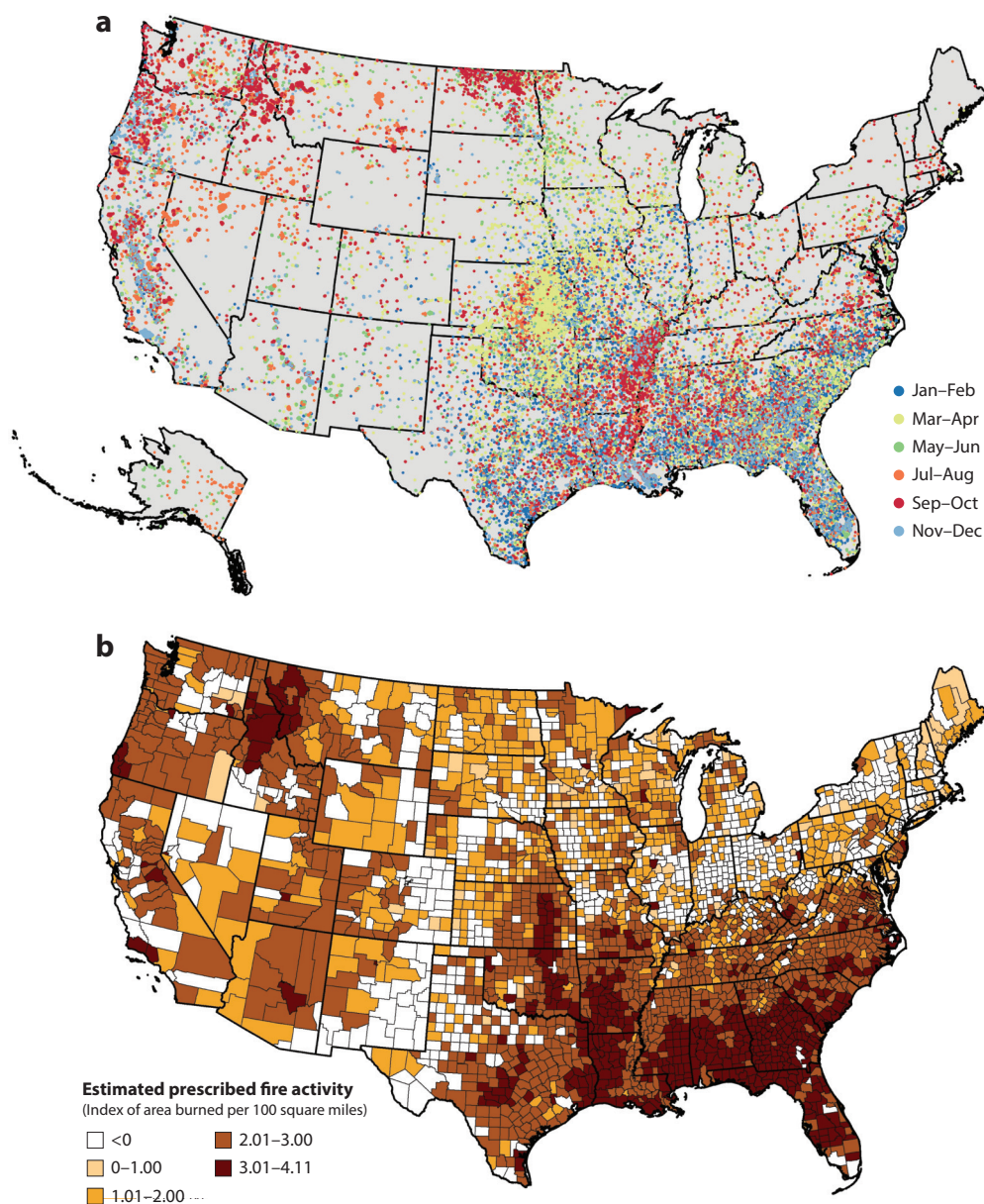


Figure 6

(a) 2017 MODIS (Moderate Resolution Imaging Spectroradiometer) fire hot spot detections color-coded by month. Panel adapted with permission from Reference 102. (b) Prescribed fire activity in the United States. Panel adapted with permission from <https://cohesivefire.nemac.org/node/63>.

information system)-based online analysis tool developed to provide predictions of the impacts of prescribed fire smoke on air quality and human health (111, 124, 125). Currently, its focus is on PB in the Southeast United States, although the air quality model domain covers the continental United States. The system uses a probabilistic, decision tree algorithm to predict the location

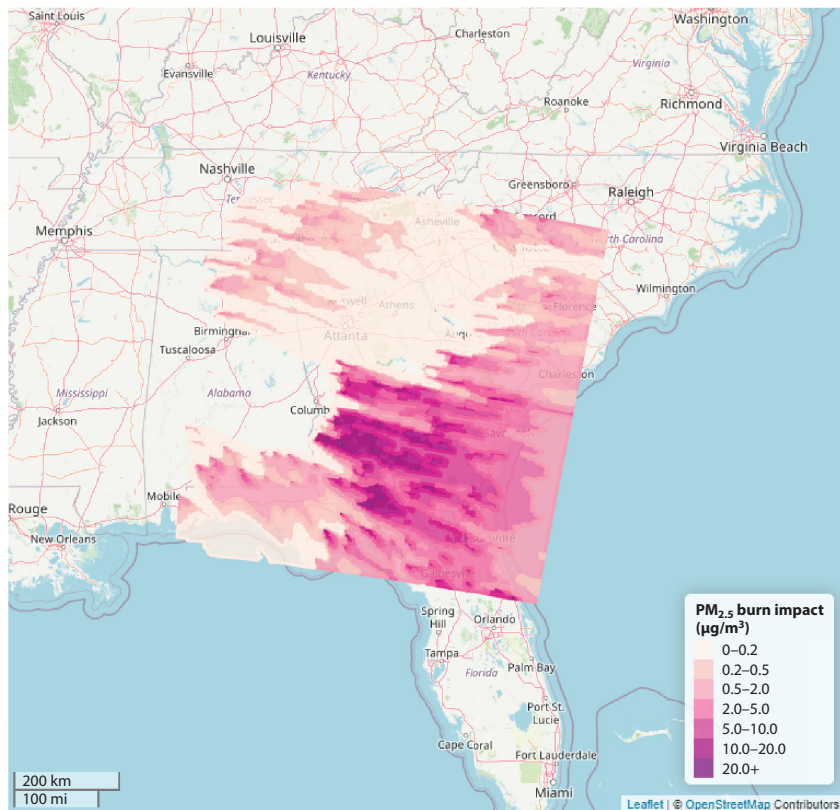


Figure 7

Prescribed burn impact forecast on January 25, 2020, by the Southern Integrated Prescribed Fire Information System (<https://sipc.ce.gatech.edu/SIPFIS/map/>). Abbreviation: PM_{2.5}, particulate matter <2.5 µm in diameter.

and size of PBs and is based in part on the Geostationary Operational Environmental Satellite (GOES), MODIS, and VIIRS retrievals of historic fire locations and sizes, as well as weather and forest conditions. Next, that information is fed into an air quality forecasting system called HiRes-X, which uses satellite observations to improve model simulations and forecast the likely impacts of PBs. (Figure 7). These forecasts are completed a day in advance to support forest and air quality managers' decisions about whether permits should be given, which are based in part on reducing human exposure. The HiRes-X system has been evaluated in part by low-cost sensors at schools and has been used for health studies by the US Forest Service and others (126).

6. SATELLITE DATA FOR DUST, AEROALLERGENS, AND SPECIALIZED SOURCES

Several other aerosols have impacts on human health and are amenable to remote sensing to support exposure assessments and related health applications. Progress on these activities depends on advances in sensing technology, the availability of ground observations, and priority applications, from exposure assessments to surveillance and early warning platforms. Dust and aeroallergens are two important examples that illustrate the opportunities and challenges of remote sensing for other specialized sources.

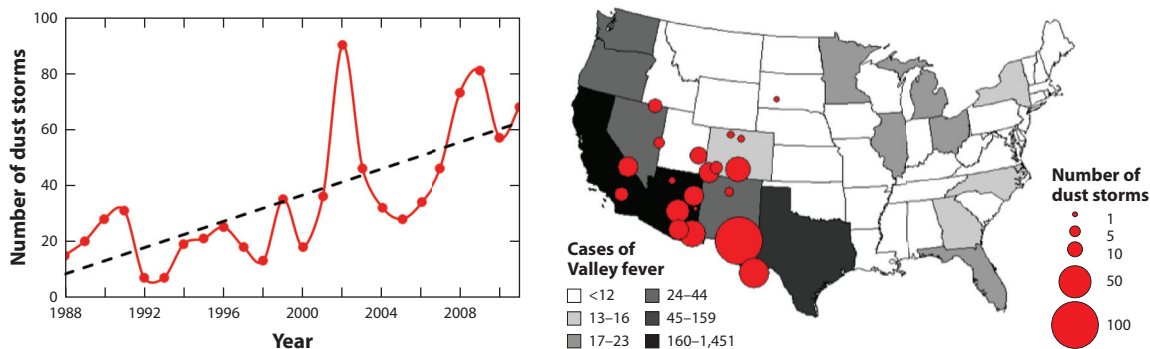


Figure 8

(Left) Frequency of dust storms increased at 12% per year between 1990 and 2011 in the United States. (Right) Dust storms (red circles) and Valley fever incidence (gray shading) in the United States. Map generated using data from Reference 130.

Dust storms are an air quality hazard that pose severe risks to public health and transportation safety in many parts of the world. Exposure to dust particles has been associated with a variety of adverse health effects, including cardiovascular mortality and heart attacks (127, 128). Dust storms also are associated with infectious diseases, such as meningitis in Africa and, in the Americas, *Coccidioidomycosis*, commonly known as Valley fever, an infection caused by inhalation of a soil-dwelling fungus (129, 130). Satellite observations are used to address several issues related to dust storms, such as reconstructing historical trends, identifying dust sources, and improving dust forecasts. For instance, Tong et al. (131) used the true color images of dust storms from the MODIS sensor aboard the Terra and Aqua satellites to train a dust detection algorithm and reconstructed long-term dust climatology over the western United States, which found that the frequency of locally originated windblown dust storms increased by 240% between 1990 and 2011 in the Southwest United States, which has high incidence rates of Valley fever infection (**Figure 8**) (130, 131).

The global remote sensing record has been widely used to identify active dust source areas (132–134). Satellite techniques and on-the-ground studies of soil and land cover characteristics of dust-emitting sites have clarified the relative frequency and intensity of dust emissions from different land use and soil types (133–135). Ginoux et al. (136) demonstrated the use of MODIS Deep Blue Level 2 (M-DB2) aerosol products to identify dust sources in West Africa. This approach identifies dust plumes by comparing dust optical depth (DOD) to a threshold value (136). An entry of DOD is found from the M-DB2 data if three criteria are met: (a) The angstrom wavelength exponent is negative ($\alpha < 0$), (b) the single-scattering albedo (ω) is smaller than 0.95, and (c) there is a positive difference in ω between 412 and 670 nm [$\omega(670) - \omega(412) > 0$]. If the determined DOD entry has a value smaller than 0.2, the satellite AOD record is considered a dust event. The abovementioned studies have revealed that dust is not emitted uniformly from across the landscape but is instead emitted from isolated hot spots in so-called preferential source areas by airborne plumes that coalesce so that a small fraction of the landscape comprises a majority of dust sources.

Aeroallergens, including pollens from wind-pollinated plants and fungal spores, are important environmental health exposures. Some pollens are highly allergenic, as are certain fungal spores. Exposure reliably leads to immune sensitization in childhood or early adulthood, and repeat exposure in sensitized individuals leads to allergic symptoms (137, 138). The most common symptom complex is allergic rhinitis, a syndrome of itchy eyes, runny nose, sneezing, fatigue, and

difficulty concentrating (139, 140). Less common but more serious is allergic asthma, a reversible airway inflammation (141). Allergic rhinitis symptoms can be managed with medications but are difficult to eliminate entirely, causing substantial morbidity and costs at the population level that are mostly related to absenteeism and presenteeism at work and school (139, 140). In severe cases, allergic asthma results in emergency department visits and hospital admissions (142–146). There appear to have been significant increases in population-level sensitization and symptomatic allergic disease over the past several decades, particularly in high-income countries and areas transitioning to industrial economies (147–150). There is also growing evidence that climate change is driving changes in pollen exposure, with significant potential impacts on human health (149, 151–154).

Information related to pollen and mold exposures is important for medical and public health activities. Pollen and mold data can support the diagnosis of allergic disease and the clarification of which allergens are driving symptoms in sensitized individuals (155). Information about pollen and mold concentrations in the atmosphere, particularly forecasts, can guide medical and behavioral management in sensitized individuals, including medication timing and exposure avoidance (156). Airborne pollens and molds can be sampled and identified by specially trained individuals, and monitoring networks are established in several parts of the world. Aeroallergen monitoring is expensive, however, requiring specialized sensors and trained identifiers who can speciate pollen samples and consistently report results. As a result, aeroallergen monitoring is limited in many locations, and the data that are available are frequently not public.

Remote sensing has long held promise as a potential source for aeroallergen data (157). Both direct and indirect methods hold promise. Aeroallergens range in size from 5 to 200 μm , and their presence can be detected remotely through the air column using different methods (158). Different approaches to measuring AOD combined with ground observations have been used to estimate pollen concentrations, generally with limited discriminatory power outside periods of high pollen concentration, but recent work suggests that micropulse lidar may hold more promise (159, 160). Indirect methods using vegetation indices such as the normalized difference vegetation index (NDVI), linked with ground observations of plant phenology and aeroallergen concentrations, also hold promise, but challenges remain (161, 162). Vegetation indices are less useful for identifying phenophase in evergreen forests and can be less reliable for estimating later-season phenology for grasses and weeds and in areas with dense vegetation (163, 164). Recent advances hold promise for using vegetation indices as phenophase correlates across a range of biomes and at large scales, however, and species distribution mapping using MODIS NDVI has potential in settings meeting certain conditions (165–167).

Remote sensing of vegetation indices and weather variables, combined with ground observations and with proxy data like web searches, has great potential for modeling aeroallergen concentrations. Skilled pollen models may help improve disease management, facilitate diagnosis, and support epidemiological analyses, including analyses regarding the contribution of climate change to the changing burden of allergic disease (168, 169). In some regions, however, taking full advantage of the potential for remote sensing to better estimate aeroallergens is likely to require additional investments in ground observations.

7. CURRENT SATELLITE CAPABILITIES AND LIMITATIONS

The previous sections survey the main applications of satellite data for air quality and health and provide a variety of specific examples of satellite data and their limitations. As we conclude this review, we step back and discuss the capabilities and limitations of satellite data more broadly. Since $\text{PM}_{2.5}$ and NO_2 have emerged as the data products most widely used for satellite-based air

quality and health applications, in this section we discuss the current and future capabilities for these two pollutants in particular.

7.1. Satellite Data for Fine Particulate Matter

From the vantage point of space, satellite remote sensing can provide information on aerosol optical properties with a wide spatial coverage, a long data record, and consistent data quality. Since the early 2000s, researchers have begun exploring quantitative methods to convert satellite data such as AOD to $\text{PM}_{2.5}$ mass concentrations to cover rural and suburban populations. AOD retrieved by instruments on polar-orbiting satellites such as MODIS and VIIRS have been extensively used in various modeling frameworks to estimate daily or long-term $\text{PM}_{2.5}$ concentrations worldwide (61, 170, 171). In particular, MAIAC is an advanced algorithm providing aerosol retrievals globally twice a day at 1-km spatial resolution (172). Many studies have shown that statistical models driven by MAIAC are able to reflect daily $\text{PM}_{2.5}$ fluctuations and intraurban exposure contrasts and extend $\text{PM}_{2.5}$ exposure estimates to regions without routine ground $\text{PM}_{2.5}$ monitoring (173–175). The long data records of satellite AOD have also allowed for the estimation of historical $\text{PM}_{2.5}$ levels where regulatory air quality monitoring networks have only been established recently (176). These desirable features have facilitated research on the health effects of both acute and chronic exposure of $\text{PM}_{2.5}$ at regional to global scales (176).

A major limitation of satellite AOD products is the large proportion (on average, 50–70%) of nonrandom missing data resulting from cloud cover and high surface brightness (e.g., snow/ice cover) (177, 178). Spatial smoothing is one way to fill in the data gaps and obtain continuous $\text{PM}_{2.5}$ predictions. For example, Kloog et al. used universal kriging with daily mean $\text{PM}_{2.5}$ surfaces and random slopes to generate fully covered $\text{PM}_{2.5}$ exposure estimates (179). Simulated AOD by atmospheric CTMs have also been used as an alternative to satellite AOD in $\text{PM}_{2.5}$ prediction (180). Statistical gap-filling models have also been developed to allow for changing relationships between AOD and meteorological factors, especially cloud fraction and snow coverage. For example, Bi et al. developed an ML-based AOD gap-filling model with snow/cloud fractions and meteorological covariates that generated spatially continuous AOD data at a 1-km resolution at a daily level, with a mean validation of $R^2 > 0.9$ (181). Gap filling has been increasingly used as a routine preprocessing step in high-resolution $\text{PM}_{2.5}$ exposure modeling.

Because polar-orbiting satellites only make one or two snapshots a day over the study region, they do not capture rapidly changing air pollution events such as dust storms and wildland fires. A major advantage of the geostationary platform is its ability to observe a given region repeatedly during daytime. For example, over the United States, fire spot, smoke mask, and AOD data, as well as aerosol detection products, are available from GOES-16 every 15 min. Aerosol retrievals from geostationary satellites have been evaluated for air quality monitoring (182–184). GOES-16 AOD data have been used together with simulated $\text{PM}_{2.5}$ concentrations from backward trajectory models to assess aerosol emission sources (185). More research is needed to develop methods to fill AOD data gaps at the hourly level.

As discussed above in various examples, surface $\text{PM}_{2.5}$ from satellite data must incorporate data from ground monitors and/or model data through fusion techniques. An example of a monitor-and-satellite-fused $\text{PM}_{2.5}$ dataset is shown in **Figure 9**, which demonstrates a visualization tool for daily, 3-km surface $\text{PM}_{2.5}$ data for the entire state of California. The $\text{PM}_{2.5}$ fields are constructed by a geostatistical method that merges EPA AirNow monitor data and NASA MODIS satellite Dark Target 3-km AOD data, using a linear regression model and a surface-smoothing model based on the inverse distance weighted method (16, 186, 187). The usage of satellite data enables the generation of this type of $\text{PM}_{2.5}$ data at near real-time frequency (i.e., within 24 h of the satellite swaths), providing timely information to facilitate the decision-making of air agencies and public

PM_{2.5} (μg/m³) in California's air basins

Daily average on May 8, 2020

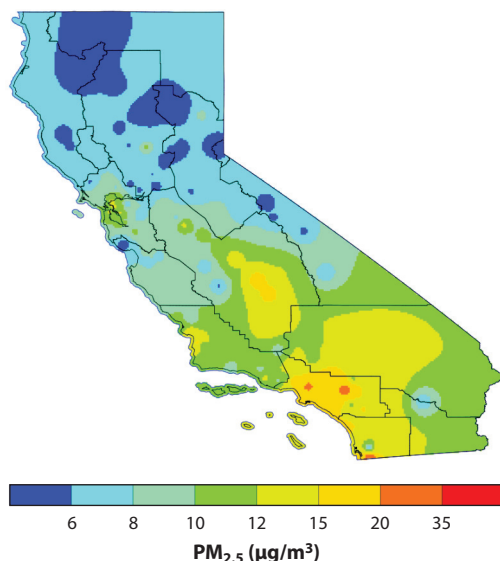


Figure 9

An example of satellite-and-monitor-fused PM_{2.5} data for the entire state of California on May 8, 2020. Daily, near-real-time, 3-km PM_{2.5} data were generated by the team of SJSU and NASA USRA, which are publicly accessible at the SJSU website (<http://www.met.sjsu.edu/weather/HAQAST/product2.html>). Maps and NetCDF files of previous days may be downloaded from archive (http://www.met.sjsu.edu/~000253930/HAQAST/PM25_CALIFORNIA_DAILY/). Figure prepared by Mohammad Al-Hamdan, Frank Freedman, and Minghui Diao. Abbreviations: NetCDF, network common data form; PM_{2.5}, particulate matter <2.5 μm in diameter; SJSU, San Jose State University; USRA, Universities Space Research Association.

health users. For health researchers that focus on the contiguous United States, a series of publicly available datasets have been documented (10), some of which are also available on a global basis.

To date, the overwhelming majority of satellite-driven exposure modeling has focused on PM_{2.5} mass concentration. A better characterization of PM_{2.5} speciation is much needed, as the spatial coverage of PM_{2.5} speciation monitors is far worse than that of PM_{2.5} mass monitors. Currently MISR and the future MAIA (discussed above in Section 2) offer information on aerosol type, which can be used to evaluate chemical speciation and associated health impacts.

7.2. Satellite Data for Nitrogen Dioxide

The newest satellite instruments are able to capture spatial heterogeneity in NO₂ levels between urban, suburban, and rural areas and can detect individual NO₂ emission sources such as power plants and industrial facilities (188). Satellite data from GOME2 (Global Ozone Monitoring Experiment 2) and SCIAMACHY (Scanning Imaging Absorption Spectrometer for Atmospheric Chartography) have been used together with datasets on land use characteristics to generate NO₂ exposure estimates at fine spatial scales (e.g., 100 m × 100 m) globally (54). This existing dataset is now being updated to leverage the long temporal record of OMI NO₂ column measurements and the high spatial resolution of tropospheric monitoring instrument (TROPOMI) NO₂ column measurements.

June 2018, clear sky NO₂

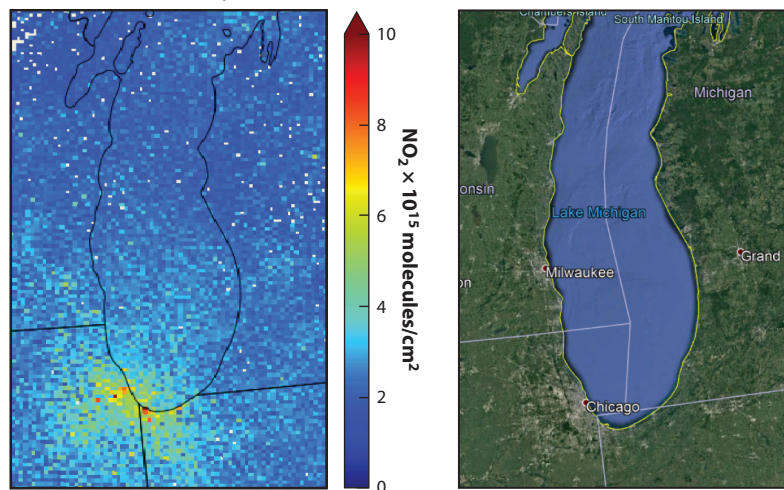


Figure 10

(*Left*) Clear sky tropospheric NO₂ columns over the Lake Michigan region aggregated onto a 4-km × 4-km grid for June 2018. (*Right*) Landsat true color image of the Lake Michigan region with major urban areas labeled. Right panel reproduced from Google Earth.

Current polar-orbiting ultraviolet–visible (UV–VIS) satellite instruments provide global, high–spatial resolution measurements of key ozone and aerosol precursor emissions at 1:00 PM local time. The TROPOMI instrument, onboard the European Space Agency’s Sentinel-5 Precursor, measures key atmospheric constituents, including ozone, NO₂, SO₂, carbon monoxide (CO), methane (CH₄), and HCHO (189). TROPOMI tropospheric NO₂ retrievals have a spatial resolution of 7 × 3.5 km² at nadir, which allows for monitoring of urban-scale NO_x emissions on monthly timescales (190). **Figure 10** shows TROPOMI clear sky tropospheric NO₂ columns aggregated onto a 4-km × 4-km grid for June 2018. A band of high–tropospheric NO₂ columns (6–10 × 10¹⁵ molecules/cm²) are observed extending from downtown Chicago, Illinois, to the southwest, as well as high–tropospheric NO₂ columns over Gary, Indiana. Elevated columns (4–6 × 10¹⁵ molecules/cm²) cover the Chicago metropolitan area and extend to the east over Lake Michigan.

The future constellation of UV–VIS satellite instruments will include geostationary, daytime measurements of ozone and aerosol precursor emissions with high spatial (3–5 km) and temporal (hourly) resolution. The Korean Geostationary Environmental Monitoring Spectrometer (GEMS) will observe emissions over East Asia; it is currently in geostationary orbit and conducting on-orbit calibration and validation activities (191). NASA’s Tropospheric Emissions: Monitoring of Pollution (TEMPO) will observe emissions over North America, and Europe’s Sentinel-4 will observe emissions over Europe using UV–VIS–near-infrared sensors (192, 193). These new geostationary UV–VIS sensors will provide measurements with high spectral, temporal, and spatial scales for retrieving trace gas concentrations relevant to air quality.

8. CONCLUSION

Health and air quality communities have grown increasingly engaged in the utilization of satellite data, and this trend is expected to continue. New generations of satellite data, including TROPOMI, TEMPO, GEMS, and Sentinel-4, will provide higher-resolution data and, for

geostationary instruments, greater temporal coverage of select world regions. In addition to data supporting new health applications, the planned NASA MAIA mission is the first satellite launched by the United States specifically focused on health research questions. From health researchers to air quality managers and from global applications to community impacts, satellite data are transforming the way air pollution exposure is evaluated.

To discuss the opportunities for evaluating air quality with new data sources, in May 2017 the American Thoracic Society, the EPA, NASA, and the National Institute of Environmental Health Sciences convened a workshop of global experts from multiple disciplines and agencies to discuss capabilities of monitoring global air quality (2). Recommendations for research and improved data use were identified during the workshop, and it was recognized that the integration of data across monitoring technology groups is necessary to maximize the effectiveness of these technologies. Rather than viewing various monitoring technologies as competing for supremacy for use in health research, it is more constructive to realize that obtaining spatially resolved estimates of short- and long-term pollution concentrations with global spatial coverage requires combining the strengths of multiple monitoring technologies.

Despite this transformation, satellite data continue to pose challenges for new user communities. Satellite data do not provide the same information as ground monitors, and they are not always appropriate to integrate with existing analysis and decision frameworks. Rather, the appropriate use of satellite data may require a rethinking of application goals—posing questions appropriate to these novel data that would not have been possible with limited ground-based monitoring. In other cases, the utilization of satellite data is only possible through data fusion techniques with models or monitors. Unfortunately, these conceptual challenges to satellite data applications compound the practical limits facing new users of satellite data: processing new data formats, selecting among a growing inventory of instruments and retrievals, and navigating the vocabulary and acronyms common in the satellite data community.

Emerging technologies for low-cost and portable monitors are expanding the spatial coverage of monitoring data (e.g., PurpleAir), including those supporting community health analysis. However, these monitors vary in their accuracy, are not considered equivalent to traditional monitors in providing exposure data, and are still found primarily in urban areas of developed countries. Although computer models have often been used to complement ground monitors, they are limited by uncertainties in model skill and required inputs. Indeed, monitoring data are often used to evaluate and improve models, so a region with few monitors may also suffer from less reliable model data. In the context of satellite data analysis, models can play an essential role linking column values measured from space with surface pollution concentrations relevant to public health. The potential for satellite data to support public health will depend on continued development of data fusion methods to link monitors, models, and space-based observations.

The NASA HAQAST effort running from 2016 to 2020 has worked to build partnerships, develop resources, and apply research to use satellite data to improve air quality and health. In late 2020, NASA announced the next generation of this team, which will advance these goals from 2021 to 2025 (<https://haqast.org>). As the health, air quality, and remote sensing communities work together with ever-improving data sources, the role of satellite data for health exposure is expected to grow rapidly in coming years.

DISCLOSURE STATEMENT

The authors are not aware of any affiliations, memberships, funding, or financial holdings that might be perceived as affecting the objectivity of this review.

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